AD 2. AERODROMES

VOCI AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VOCI - COCHIN INTERNATIONAL AIRPORT, NEDUMBASSERY

VOCI AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	100913N 0762426E 48M North of Taxiway C and 590M East of TWY H	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	045 DEG, 28KM from Cochin South Railway Station	
3	Aerodrome elevation and reference temperature	30 FT / 29.6 DEG C	
4	Magnetic variation, date of information and annual change	2.12 DEG W (2010) /0.0583 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Managing Director Cochin International Airport Limited, (CIAL), Kochi Airport P.O, Cochin – 683111 Kerala, India,	
		Telephone:	(91) 4842610125 (91) 4842610115
		Fax:	(91) 4842610009
		AFS:	VOCIYOYX
		Email:	md@cial.aero
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	Website: www.cial.aero	

VOCI AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	H24
2	Custom and immigration	H24
3	Health and sanitation	Available
4	AIS briefing office	H24
5	ATS reporting office (ARO)	H24
6	MET Briefing office	H24
7	Air Traffic Service	H24
8	Fuelling	H24
9	Handling	H24
10	Security	H24
11	De-icing	Not applicable

12	Remarks	1. The approved hourly RWY traffic handling capacity as follows: Peak Capacity (When ASR Available) Maximum number of arrivals and departures -29 Maximum number of arrivals only -15 Maximum number of departures only -14 2. AD Reference Code: 4E 3. Aerodrome License No. AL / PUBLIC / 003 Category - Public use (See NOTAM for the validity of license) 4. All Non-Scheduled / General Aviation movements will be subject to positive approval from Airport Operations Control Centre (AOCC). Email : aocc@cial.aero Mob : 9539273172

VOCI AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	Cargo Terminal Management-CIAL Handling Capacity: International-350MT per day Domestic -60 MT per day.
2	Fuel and Oil types	JET A1
3	Fuelling facilities and capacity	Hydrant fuelling system, 4000 litre/minute Fuel Service Providers – BPCL, IOCL, HPCL Bowser fuelling capacity BPCL – 12KL/ IOCL– 11KL
4	De-icing facilities	Not Available
5	Hangar space for visiting aircraft	Code C - 2 Nos.
6	Repair facilities for visiting aircraft	Transit Checks, Layover inspection and Daily checks for A320 & A330 series.
7	Remarks	Ground Handling Service provided by Air AIASL, BWFS & CELEBI

VOCI AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	Near the AD and in the city
2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city
3	Transportation possibilities	Buses, Pre-paid taxis and car hire from the AD. Two Railway stations at 6 & 12 KM distance. Metro Station at 12 KM distance.
4	Medical Facilities	Exclusive Medical Intervention facility available at both International and Domestic Terminals. Hospitals at 5 KM and in city.
5	Bank and post office at or in the vicinity of aerodrome	Banks: At AD. Post office: At AD.
6	Tourist office	At AD: +91-484-2611308/9, City Office: +91-484-2353234
7	Remarks	NIL

VOCI AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-9
2	Rescue equipment	Available

3	Capability for removal of disabled aircraft	1. The critical aircraft for the airport is B747-400. (Disabled Aircraft recovery capability is limited to Code C) 2. Primary responsibility for removal of disabled aircraft rests with the concerned Aircraft Operator , in coordination with the Airport Operator CIAL. 3. The airlines may contact 'Aerodrome Nodal Officer / Alternate Nodal Officer for Disabled Aircraft Removal Operations' for coordinating the Disabled Aircraft Recovery Management. 4. Contact Details of Aerodrome Nodal Officer and Escalation Nodal Officer for Disabled Aircraft Removal Operations: 1) Nodal Officer : Mr Sojan Koshy Phone : 9539701537 E-mail : sojan@cial.aero 2) Escalation Nodal Officer : Mr Abraham Joseph Phone : 9539701503 E-mail : abrahamj@cial.aero
4	Remarks	NIL

VOCI AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	Sweeping machine available
2	Clearance priorities	Runway 09/27 and associated taxiways
3	Remarks	No snow fall

VOCI AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Refer Aircraft Parking/Docking Chart & AD2.23
2	Designation, width, surface and strength of taxiways	Refer Aircraft Parking/Docking Chart & AD2.23
3	Location and elevation of altimeter checkpoints	Location Elevation Apron 1 26FT Apron 2 28FT Apron 4 28FT Apron 5 26FT Apron 7 26FT
4	Location of VOR checkpoints	RWY Holding Positions at: Taxiway C2: 086 DEG, 1.1 NM Taxiway C7: 088 DEG, 2.5 NM
5	Position of INS checkpoints	

6	Remarks	i. Location of TWY C2 : 2680M from beginning RWY27, North of RWY 09/27 ii. Location of RET C3 : 1980 M from beginning of RWY 27, North of RWY 09/27. iii. Location of TWY C4 : 1550M from beginning RWY27, North of RWY 09/27 iv. Location of RET C5: 2300 M from beginning of RWY 09, North of RWY 09/27. v. Location of TWY C6: 3100M from beginning RWY09, North of RWY 09/27. vi. Hangar Apron Area - 31, 344 sq m vii. Aprons 1, 2 and 4 are power-in push back viii. Apron 5 - Power in and Tow/Power out. Power out from stands for aircraft up to Code B only ix. Apron 7 - Power in. Power Out from stands permitted after manual repositioning. x. TWY H-Connecting TWY C and Hangar Apron.

VOCI AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	1. Taxiing guidance signs provided at all intersections with TWY & RWY and at all Holding Positions. 2. Aircraft stand numbers, apron guidelines, taxiway guidelines marked. 3. Apron edge and aircraft stand marking available. 4. A-VDGS provided for stand nos. 1R, 2L, 2R, 3 to 6, 19-23R. 5. Stand manoeuvring guidance lights available for stands 1R,2L,2R,3,4,5,6,19,20,21,22,23,23L,23R.
2	Runway and taxiway markings and lights	RWY Markings: Designation, THR, TDZ, Centre-line. Aiming Point, Side stripe, Ends. RWY Lights: Edge, THR, END, TDZ Lights for RWY 27 & Centreline. TWY Marking: Edge, Centreline, Enhanced Centreline, RWY holding positions, Intermediate Holding Positions TWY Lights: Edge: C1, C2, C3, C4, C5,C6, C7, A, B, C, D, D1, E, H, H1& P. Centreline:C, C1, C2, C3, C4,C5, C6, C7, A, B, E, F, G, J. Taxilane Marking: Centreline, Outer Edge for J1, L, L1 & N Taxilane Lights: Edge: J1 Centreline: J1, L, L1, N & N1
3	Stop bars (if any)	Not available.
4	Remarks	1. Apron Edge and Aircraft stand markings available. 2. Follow me Sevice provided when required. 3. Signage – Illuminated

VOCI AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation	Marking/LGT	Remarks
In circling area and at AD	BUILDING	100706.4N 0762422.5E	234.849 FT	NIL	Concrete Residential Flat

VOCI AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	Cochin Airport MET office.
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H-24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Thiruvananthapuram (VOTV) Validity 9 hours 8 times a day and TAF of validity 30 HRS is issued 4 times a day
4	Availability of the trend forecast for the aerodrome and interval of issuance	METAR provided every 30Min and SPECI which is issued as and when required
5	Information on how briefing and/or consultation is provided	Personal ConsultationChart form of documentation received from RAFC / MO Chennai and ROFOR for MO Trivandrum provided and also through the On-Line Briefing System (OLBS) of IMD.
6	Types of flight documentation supplied and language(s) used in flight documentation	ChartsAbbreviated Plain language English
7	Charts and other information displayed or available for briefing or consultation	TAFs and Current Weather reports, Aerodrome warnings of VOCI and other airports provided H24. SIGWX, SIGMET, WINTEMP charts from WAFC, RADAR and Satellite images are also available.
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	METAR/ AERODROME Warning display system Online briefing system: https://olbs.amsschennai.gov.in/nsweb/FlightBriefing/ https://olbs.amssdelhi.gov.in/ IMD website: https://mausam.imd.gov.in/ E mail: ams.chn@imd.gov.in/ imdkochiairport@gmail.com Phone: 0484 2610204, 0484 2610208
9	The air traffic services unit(s) provided with meteorological information	VOCI ATC/COM/MET Briefing/Cochin TWR/Cochin APP Units
10	Additional information, e.g. concerning any limitation of service.	i. Digital CWIS for RWY 27 provided at Aerodrome Control Unit and Approach Control Unit; DWIE for RWY 09 available at Aerodrome Control Unit only. ii. Transmissometer RWY27 (Drishti) available. iii. Laser Ceilometer RWY 27 available. iv. Limitation of Service: AVRA not available for RWY 09. Manual RVR being provided on requirement basis. v. automated RVR sensor forward scatter Meter is AVBL for RWY27.

VOCI AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCR) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
09	87.64 DEG	3400 x 45 M	850/F/A/W/T Asphalt	THR: 100901.97N 0762306.00E RWY END: 100906.06N 0762457.67E
27	267.64 DEG	3400 x 45 M	850/F/A/W/T Asphalt	THR: 100906.06N 0762457.67E RWY END: 100901.97N 0762306.00E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 28.0FT TDZ: 28.0FT	0.08%	NIL	NIL	3520 x 300 M
THR: 30.0FT TDZ: 30.0FT	0.07%	NIL	NIL	3520 x 300 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
240M x 90M	NIL	AVBL	1. The Strength (PCR) of first 100m from beginning of RWY 09 is 950/R/A/W/T
240M x 90M	NIL	AVBL	1. The Strength (PCR) of first 100m from beginning of RWY 09 is 950/R/A/W/T

VOCI AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
09	3400	3400	3400	3400	
27	3400	3400	3400	3400	

VOCI AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	**2**	**3**	**4**	**5**
09	CAT I 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (66.04FT)	NIL
27	CAT I 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (66.66FT)	900 M

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
3400 M 30 M LIH White	3400 M 60 M White LIH	Red	NIL	1. RWY Centreline Lights from physical beginning RWY 09 up to 900M before RWY End – variable White; between 900M and 300M before RWY 09 End – alternate variable White and Red; between last 300M toRWY 09 End – Red 2. RWY Edge Lights from physical beginning RWY 09 up to 600M before RWY 09 End) – Variable White; between last 600M to RWY 09 End – Yellow. 3. CAT1 APP lights Barrette type with Seq. flasher. 4. Rapid Exit TWY Indicator Lights (RETIL) installed at RWY 09 for TWY C5.
3400 M 30 M LIH White	3400 M 60 M White LIH	Red	NIL	1. RWY Centreline Lights from physical beginning RWY 27 up to 900M before RWY End – variable White; between 900M to 300M before RWY 27 End – alternate variable White and Red; between last 300M and RWY 27 End – Red 2. RWY Edge Lights from physical beginning RWY 27 up to 600M before RWY 27 End) – Variable White; between last 600M to RWY 27 End – Yellow. 3. Rapid Exit TWY Indicator Lights (RETIL) installed at RWY 27 for TWY C3 - Yellow. 4. Touchdown zone lights – barrettes spaced 30M till 900M from THR –Variable White. 5. CAT-I Approach lights -Calvert type with five Crossbars over 900 M. 6. Approach Side row lights - barrettes spaced 30M till 270M from THR RWY 27 - Red.

VOCI AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	At Tower building H-24 RPM 15 per minute
		IBN	NIL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	Located at 1950M from RWY09 THR and 100M North of RWY Centreline, lighted.
		Anemometer	300M from THR RWY 09 & 27; 120M South of RWY Central Line
3	Taxiway edge and taxiway centre line lights;	Edge	TWY C1, C2, C3, C4, C5, C6,C7 A, B, C, D, E, H, H1, P, Joining portion of TWY F & G with TWY C - Elevated Blue lights. Taxilane J1 - Elevated Blue lights on Southern side.
		Centre Line	TWY C1, C2, C3, C4, C5, C6, C7, A, B, C, E, F, G, J Taxi Lanes L, L1, J1, N & N1 Inset bidirectional Green lights.
4	Secondary power supply including switch-over time;	Secondary power supply to all AGL and NAV AID Systems at AD. Switch over time - 15sec.	
5	Remarks	Runway Guard Lights Available at C1, C2, C3, C4, C5 WDI: Illuminated WDI at RWY 09 &27. Non- illuminated WDI at Fire Station Approach Road.	

VOCI AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	No separate landing area for Helicopters
2	TLOF and/or FATO area elevation:	Not applicable
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not applicable
4	True bearings of FATO;	Not applicable
5	Declared distances available	Not applicable
6	Approach and FATO lighting;	Not applicable
7	Remarks	Helicopters to land on Runway 09/27

VOCI AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area of radius 15 NM centered at 100913N 0762426E excluding portions of Cochin (Navy) VFR Flying Zone and Restricted Area VOR 191-A.
2	Vertical limits	5500 FT
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Cochin Tower / Cochin Approach Cochin Radar, English
5	Transition altitude	11000 FT
6	Hours of applicability	H24
7	Remarks	Vertical Limit- 5500 FT AMSL

VOCI AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available	Logon address, as appropriate	Hours of operation	Remarks
1	2	3	4	5	6	7
APP	Cochin Approach Cochin Radar	119.750 MHZ			H24	Emergency frequency 121.50 MHz with TWR, APP, SMC and ACC
APP	Cochin Approach Cochin Radar	133.150 MHZ			H24	Standby Frequency
TWR	Cochin Tower	118.800 MHZ			H24	Emergency frequency 121.50 MHz with TWR, APP, SMC and ACC
TWR	Cochin Tower	119.400 MHZ			H24	Standby Frequency
ATIS	Cochin Information	126.200 MHZ			H24	Emergency frequency 121.50 MHz with TWR, APP, SMC and ACC
ACC	Cochin Control	120.775 MHZ			H24	(Main) Emergency frequency 121.50 MHz with TWR, APP, SMC and ACC
ACC	Cochin Control	133.150 MHZ			H24	(Standby) Emergency frequency 121.50 MHz with TWR, APP, SMC and ACC
SMC	Cochin Ground	119.400 MHZ			H24	Standby Frequency
SMC	Cochin Ground	121.750 MHZ			H24	.Emergency frequency 121.50 MHz with TWR, APP, SMC and ACC

VOCI AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 27 CAT I	ICIL	110.300 MHz	H24
LOC 09 CAT I	ICNB	111.100 MHz	H24
GP 27	ICIL	335.000 MHz	H24

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
GP 09		331.700 MHz	H24
DME ILS 27	ICIL	CH72X	H24
DME ILS 09	ICNB	CH48X	H24
DVOR/DME	CIA	113.500 MHz CH82X	H24
DVOR/DME	CIB	117.300 MHz CH120X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
100900.8N 0762253.3E			
100906.4N 0762508.0E			
100901.7N 0762447.3E			
100858.4N 0762317.7E			
100901.8N 0762447.3E	40 FT		Low power
100858.4N 0762317.7E	42 FT		Low power
100901.8N 0762221.3E	46 FT		
100700.8N 0764050.1E	155 FT		

VOCI AD 2.20 LOCAL AERODROME REGULATIONS

i. Hot Spots declared - The point where Rapid Exit TWY C3 meets TWY C & point where TWY F meets Taxilanes L and L1.

ii. An aircraft vacating RWY via Rapid Exit TWY C3 or TWY C5 is not simultaneously clear of RWY and parallel TWY C.

iii. An aircraft vacating RWY via TWY C3 can continue taxi to turn left on parallel TWY C, hold short of TWY B and wait for further instructions.

iv. Aircraft not permitted to enter TWY C3 or TWY C5 from parallel TWY C.

v. Simultaneous use of TWY E and TWY C3 is permitted.

vi. Aircraft up to Code C, vacating RWY via TWY C1, C2, C4, C6 or C7 is simultaneously clear of RWY and parallel TWY C.

vii. Aircraft up to Code C Holding at Runway Holding Positions on TWY C1, C2, C4 and C6 is clear of aircraft taxiing via TWY C.

viii. 180 Deg turn on Turn pads permitted for aircraft up to Code C only.

ix. 180 Deg turn on Runway permitted for aircraft having MTOW up to 25 T only.

VOCI AD 2.21 NOISE ABATEMENT PROCEDURES

Consistent with safety of aircraft operations and in consideration of high intensity runway operations, pilots should minimise the use of reverse thrust after landing to reduce disturbance in areas adjacent to the aerodrome.

VOCI AD 2.22 FLIGHT PROCEDURES

I. ADS-B based ATS Surveillance services are provided to suitably equipped aircraft on opportunity basis within the coverage area of Cochin ADS-B sensors in the jurisdiction of Cochin Area Control.

ADS-B based ATS Surveillance services are also provided to suitably equipped aircraft on opportunity basis within the coverage area of Cochin ADS-B sensors in the jurisdiction of Cochin Approach control.

ADS-B equipped aircraft flying within Cochin Area Control and Approach Control to comply with relevant provisions.

II. Low Visibility Procedure, Cochin International Airport, Cochin (VOCI)

1. BACKGROUND

1.1 As per Civil Aviation Requirements Section 8 Series C Part 1 Subsection 5.3, an operator must enforce Low Visibility Procedures (LVP) to conduct take-offs with Runway Visual Range (RVR)/visibility less than standard Category I conditions of 550m RVR/800m visibility. This necessitates LVP for permitting departures in visibility/RVR less than 800m/550m at airports without CAT II and CAT III operations.

1.2 As per Civil Aviation Requirements Section 4 Series B Part 1 Aerodrome Design and Operations and ICAO Annex 14, runway centreline lights are required for take-off in RVR below 400m. Cochin International Airport has developed LVP to accommodate departures in visibility/RVR less than 800m/550m from RWY 09 and RWY 27. Runway centreline lights are available, permitting take-off in visibility/RVR below 800m/550m. Scheduled operators authorized by DGCA for low visibility take-offs in less than 400m RVR may depart when LVP is implemented.

2. DEFINITIONS

• Low Visibility Procedures (LVP): Procedures ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

• Manoeuvring Area: The aerodrome area used for take-off, landing, and taxiing, excluding aprons.

• Movement Area: The area used for take-off, landing, and taxiing, including the manoeuvring area and aprons.

• Low Visibility Take Off (LVTO): A take-off on a runway where RVR is less than 400m.

• Runway Visual Range (RVR): The distance over which a pilot on the runway centreline can see runway surface markings or lights.

• Aerodrome Operating Minima: The usability limits of an aerodrome for take-off, landing in 2D and 3D instrument approach operations.

• Visibility: The greater distance between the visibility of a black object against a bright background or lights of 1000 candelas against an unlit background.

3. GENERAL

3.1 LVP incorporates measures to mitigate runway incursions and defines operational restrictions to ensure safe airside operations considering available aerodrome facilities.

4. MINIMUM REQUIREMENTS

4.1 The following aeronautical ground lights and RVR equipment must be serviceable to support LVP:

• Runway edge lights for RWY 09/27

• Runway end lights for RWY 09/27

• Runway centreline lights for RWY 09/27

• Runway centreline markings for RWY 09/27

• Automated RVR for RWY 09/27

• Taxiway centreline lights for all taxiways except H, H1 & D1

4.2 Additionally, real-time visibility data availability and standby power supply ensuring a switch over time of one second for runway edge and end lights supported by UPS/DG Set is required. All AGL are supported by UPS to ensure changeover time of less than one second.

Table 1: Effect of Unserviceability of AGL on LVTO

Aeronautical Ground Lighting Facility	**Un-serviceability**	**Restrictions**
Runway Edge lights	a) More than 5% of all lights are unserviceable OR b) Any two or more consecutive lights unserviceable	LVTO Operations can come till 300 m if centre line lights and RWY End Lights are available with real time RVR.
Runway End lights	a) More than 25% of lights unserviceable OR b) Any two or more consecutive lights unserviceable	LVP operations will be suspended.
Runway Centre line Lights	a) More than 5% of lights unserviceable OR b) Any two or more consecutive lights unserviceable	LVTO only up to 300 M provided RWY Edge lights and RWY End Lights are available along with real time RVR
Standby Generators/ UPS	Generator/ UPS unserviceable	LVP operations will be suspended.

4.3 Unserviceability of Aeronautical Ground Lights/Equipment after Implementation of LVP:

4.3.1 If any light/equipment listed in Section 4.1 becomes unserviceable or fails to meet the standard as envisaged in Table 1 during LVP, TWR/SMC shall inform the aircraft accordingly, suspend LVP and information to that effect included in the ATIS broadcast.

5. SAFEGUARDING PROCEDURES

5.1 These procedures ensure that ground services and facilities are prepared for low visibility operations before implementing LVP. Duty Officer Tower AAI, in coordination with DM Operations CIAL, initiates safeguarding procedures based on RVR status and MET department lookout.

5.2 Safeguarding Procedures Initiation:

• Visibility/RVR less than 1200m and forecast to deteriorate to 800m/550m or less.

• Cloud ceiling less than 400ft and forecast to fall to 200ft or less.

5.3 Safeguarding Procedures Include:

• Positioning 1 CFT each for RWY 09 and 27.

• Stopping maintenance works on the manoeuvring area and associated strip areas.

• Inspecting Aeronautical ground lights during the hour preceding LVP implementation and before departures during LVP.

• Ensuring secondary power supply availability for a maximum one-second changeover time for RWY Edge and End lights supported by UPS/DG Set.

6. LOW VISIBILITY PROCEDURES

6.1 Implementation:

6.1.1 The Duty MET Officer informs the Tower Supervisor/Duty Officer Tower when visibility/RVR reduces to 800m/550m or below and/or cloud ceiling is 200ft or below. ATCO coordinates with DM Operations and ARFF to confirm completion of safeguarding procedures. Once confirmed, the Tower Supervisor/Duty Officer Tower implements LVP with prior WSO approval.

Table 2:

RVR/Visibility for Take-off (Commercial Transport Aeroplanes)

Facilities	**RVR / VIS (Refer Note 1) Cat A, B, C, & D**
Adequate Visual Reference (Day Only) (Refer Note 2)	500m
Runway edge lights or Runway centreline markings (Refer Note 3)	400m
Runway edge lights and Runway centreline markings (Refer Note 3)	300m
Runway edge lights and Runway centreline lights	200m

Note 1 – The TDZ RVR/VIS may be assessed by the pilot.

Note 2 - Adequate Visual reference means, that a pilot is able to continuously identify the take-off surface and maintain directional control.

Note 3 - For night operations at least runway edge lights or centre line lights and runway end lights are available.

6.2 ACTION BY VARIOUS UNITS DURING LVP:

6.2.1 MET Office:

• Issue “Outlook for Low Visibility” when RVR is less than 1200m or visibility is forecast to deteriorate to 800m or less.

• Ensure Transmissometer serviceability and provide direct reading of TDZ RVR observations.

• Deploy manpower for manual RVR if Transmissometer is unserviceable or if LVTO is to be conducted on a runway without a Transmissometer.

• Provide Leader service to obtain manual RVR and inform Tower Supervisor/Duty Officer Tower of any visibility/RVR changes.

6.2.2 TWR/SMC Controller - ATC:

• Implement LVP with WSO approval.

• Stop all ground runs on the manoeuvring area except idle power run-up on stands after coordinating with DM Operations CIAL.

• Allow only vehicles with serviceable R/T on the manoeuvring area; vehicles without R/T may operate if escorted by a Follow Me Vehicle.

• Permit departures only from the beginning of the Runway-In-Use.

• Coordinate safeguarding procedures and declare LVP in place when visibility/ RVR goes below 800m/550m.

• Stop flight operations when RVR falls below 300m and inform WSO.

6.2.2.1 Taxi and Take-off Instructions:

• Allow only one push-back/start-up at a time per apron, unless their taxi tracks do not conflict.

• Confirm the positions of preceding aircraft for consecutive taxiing departures and clear succeeding traffic accordingly.

• Departing aircraft at the holding point may enter the runway only if no landing aircraft is within 8NM from touchdown.

• Check airline operator/aircraft approval for departure in RVR less than 400 m before push-back.

6.2.3 DM Operations CIAL:

• Conduct runway inspection to ensure serviceability of RWY Edge, Centreline, and End lights.

• Inspect AGL before departures during LVP and inform ATC of any non-compliance.

6.2.4 APRON Control / AOCC:

• Ensure towing of aircraft under escort of Follow Me vehicles.

• Provide Follow Me service to non-scheduled aircraft and restrict the number of vehicles operating on the manoeuvring area.

• Do not permit high power run-up of aircraft in the movement area.

• Follow safety protocols for vehicle movement on the apron.

6.2.5 Duty Manager ARFF:

• Position 1 CFT each for RWY 09 and 27 after obtaining permission from ATC.

• Inspect runway and taxiways for compliance.

6.2.6 In-Charge - Electrical:

• Continuously monitor main and standby power supply to ensure change over time of maximum one second for critical AGL during LVP.

6.2.7 Airlines:

• Pilots must ensure RVR is within minima for LVTO while requesting push- back/start-up.

6.2.8 Ground Handling Agencies:

• Activate LVP protocols and coordinate with DM AOCC.

• Enhance communication and coordination, ensure serviceability of obstruction lights on equipment, minimize ground movement, ensure availability of retroreflective markings on equipment, ensure staff wear PPE, strictly adhere to pushback and towing procedures, pre-arrival and departure coordination, conduct regular safety briefings, follow rigorous training and adherence to protocols and prepare for emergencies.

6.2.9 Fuelling & Catering Agencies:

• Follow strict safety protocols, use appropriate lighting, reduce speed and manoeuvre cautiously, maintain enhanced communication, adhere strictly to procedures, and conduct safety checks.

7. TERMINATION OF LOW VISIBILITY PROCEDURES

7.1 When visibility/RVR improves to 800m.

VOCI AD 2.23 ADDITIONAL INFORMATION

1.ADS-B at Cochin Airport.

Facility	Frequency	Latitude	Longitude
ADS-B	1090Mhz	10 Deg 9 Min 22.32Sec	76 Deg 24 Min 1.8Sec

2. Details of Taxiways

Taxiway, Width, Surface and Strength

DESIGNATOR	WIDTH	SURFACE	SHOULDER	STRENGTH(PCR)	AIRCRAFT UPTO
A	23M	Asphalt	10.5M	850 F/A/W/T	Code E
B	23M	Asphalt	10.5M	850 F/A/W/T	Code E
C	23M	Asphalt	10.5M	850 F/A/W/T	Code E
C1	23M	Concrete	10.5M	FM RWY 78.4M FM TWY 32M-850 F/A/W/T	Code E
		Asphalt		REMAINING PORTION 47.6M OF TWYC1- 990 R/A/W/T	
C2	23M	Asphalt	10.5M	850 F/A/W/T	Code E
C3	23M	Asphalt	10.5M	850 F/A/W/T	Code E
C4	23M	Asphalt	10.5M	850 F/A/W/T	Code E
C5	23M	Asphalt	10.5M	FM RWY 92.2M FM TWY 55.5M- 850 F/A/W/T	Code E
		Concrete		REMAINING PORTION 159.1M OF TWYC5-990 R/A/W/T	
C6	23M	Asphalt	10.5M	FM RWY 47.5M FM TWY 28M- 850 F/A/W/T	Code E
		Concrete		REMAINING PORTION 82.5M OF TWYC6- 990 R/A/W/T	
C7	23M	Asphalt	10.5M	FIRST 80.4M LAST 31.6M-850 F/A/W/T	Code E
		Concrete		REMAING PORTION 46M OF TWYC7-990 R/A/W/T	
D	23M	Asphalt	10.5M	850 F/A/W/T	Code E
D1	15M	Asphalt	5.0M	850 F/A/W/T	Code C
E	23M	Asphalt	10.5M	850 F/A/W/T	Code E
H	23M	Asphalt	10.5M	850 F/A/W/T	Code E
H1	10.5M	Asphalt	12.0M	850 F/A/W/T	Code B

F	23M	Asphalt	10.5M	FM TWY C 86.5M-850 F/A/W/T	Code E
		Concrete		REMAINING PORTION 214.3M OF TWY F-990 R/A/W/T	
G	23M	Asphalt	10.5M	FM TWYC 86.5M-850 F/A/W/T	Code E
		Concrete		REMAINING PORTION 214.3M OF TWY G-990 R/A/W/T	
J	23M	Concrete	10.5M	990/R/A/W/T	Code E
P	10.5M	Asphalt	5.0M	850 F/A/W/T	Code B

3. Details of Taxilanes

DESIGNATOR	WIDTH	SURFACE	STRENGTH(PCR)	AIRCRAFT UPTO
L	23M	Concrete	990/R/A/W/T	Code E
N	23M	Concrete	990/R/A/W/T	Code C
L1	23M	Concrete	990/R/A/W/T	Code E
N1	22M	Concrete	990/R/A/W/T	Code C
J1	23M	Concrete	990/R/A/W/T	Code E

4.Details of Apron & Aircraft Stands

Apron 1				Elevation: 26 FT
Aircraft Stand No	**Surface**	**PCR**	**Coordinates**	**Suitability**
1	Concrete	990/R/A/W/T	100915.82N 0762318.64E	Code E
1L	Concrete	990/R/A/W/T	100915.38N 0762317.96E	Code C
1R	Concrete	990/R/A/W/T	100915.44N 0762319.63E	Code C
2L	Concrete	960/R/A/W/T	100915.53N 0762321.34E	Code C
2R	Concrete	960/R/A/W/T	100915.53N 0762322.94E	Code C
3	Concrete	960/R/A/W/T	100916.12N 0762324.96E	Code C
4	Concrete	960/R/A/W/T	100916.20N 0762327.42E	Code C
5	Concrete	960/R/A/W/T	100916.09N 0762329.63E	Code C

6	Concrete	960/R/A/W/T	100916.11N 0762331.43E	Code C
7	Concrete	990/R/A/W/T	100916.21N 0762332.96E	Code C
8	Concrete	980/R/A/W/T	100916.25N 0762334.24E	Wingspan upto 32 M
9	Concrete	980/R/A/W/T	100916.30N 0762335.39E	Wingspan upto 25 M
10	Concrete	980/R/A/W/T	100916.34N 0762336.55E	Code C
10A	Concrete	980/R/A/W/T	100915.38N 0762337.69E	Wingspan upto 17 M
11	Concrete	980/R/A/W/T	100916.42N 0762338.72E	Code C
12	Concrete	980/R/A/W/T	100916.49N 0762340.59E	Code C
13	Concrete	980/R/A/W/T	100916.56N 0762342.44E	Code C
14	Concrete	980/R/A/W/T	100916.61N 0762343.87E	Code C
15	Concrete	990/R/A/W/T	100916.84N 0762346.49E	Code C
16	Concrete	990/R/A/W/T	100916.90N 0762347.92E	Code C

1. Aerobridge available for Stands 1R, 2L, 2R, 3, 4, 5, 6.
2. A-VDGS available for stands 1R,2L,2R, 3, 4, 5, 6. Manual Marshalling for all remaining stands.
3. Hydrant fuelling facility available for stands 1L - 16 except 10A.
4. Push back & Taxi procedure:
   - 4.1 Push back in the same direction, shall be given with separation of 02 clear stands.
   - 4.2 Stand 10A is not considered for this clearance count.
   - 4.3 Stands 1L, 1R, 2L & 2R are considered as independent stands for the purpose of clearance count.
   - 4.4 If aircraft is pushed back from stand 1, next push back in same direction to be given from stand 3.
   - 4.5 When aircraft at stand 1L or 1R is in pushback position facing East, Stand 1L & 1 is not clear for an aircraft to taxi -in
   - 4.6 Aircraft pushing back facing West from stands 15 & 16 to be towed till Tug Disconnect Point -1 (TDP-1) abeam Stand 15, to be clear of TWY F
   - 4.7 Engine start up permitted upon completion of push back.

Apron 2				**Elevation: 28 FT**
Aircraft Stand No	**Surface**	**PCR**	**Coordinates**	**Suitability**
19	Concrete	990/R/A/W/T	100924.40N 0762342.51E	Code E
20	Concrete	990/R/A/W/T	100924.49N 0762344.89E	Code E

21	Concrete	990/R/A/W/T	100924.59N 0762347.68E	Code E
22L	Concrete	990/R/A/W/T	100924.67N 0762350.06E	Code C
22	Concrete	990/R/A/W/T	100923.76N 0762349.62E	Code E
22R	Concrete	990/R/A/W/T	100924.79N 0762350.36E	Code C
23L	Concrete	990/R/A/W/T	100924.39N 0762353.23E	Code C
23	Concrete	990/R/A/W/T	100923.86N 0762352.64E	Code E
23R	Concrete	990/R/A/W/T	100924.91N 0762353.44E	Code C

1. Aerobridge available for all stands.
2. AVDGS available for all stands.
3. Hydrant fuelling facility available for all stands.
4. Push back & Taxi procedure:
   - 4.1 Push back in the same direction, shall be given with separation of 02 clear stands.
   - 4.2 Aircraft from Stand 19 to be pushed back till Tug Disconnect Point-2, for another aircraft to Taxi-in to stand 19.
   - 4.3 Stands 22L/22R and 23L/23R are considered as 22 and 23 respectively for the purpose of pushback calculation.
   - 4.4 Aircraft from Stand 23 & 23R, pushed back facing West, to be towed till Tug Disconnect Point-3.
   - 4.5 Aircraft taxiing via Taxiway G and N, is clear of aircraft upto Code C facing west and starting engines at Tug Disconnect Point - 3.
   - 4.6 Taxi via Taxiway G and N not permitted when a code D/E aircraft is pushing back facing west / starting on Taxilane L.
   - 4.7 Simultaneous push back / simultaneous entry or simultaneous pushback & entry not permitted from/to stands 22R and 23L.
   - 4. 8 Engine start up permitted upon completion of push back/tow.
5. Aircraft entering stands 19-21 to exercise caution due proximity to Equipment parking area on South.

Apron 4				Elevation: 28 FT
Aircraft Stand No	Surface	PCR	Coordinates	Suitability
35	Concrete	990/R/A/W/T	100926.69N 0762401.37E	Code C
36	Concrete	990/R/A/W/T	100925.37N 0762401.42E	Code C

1. Hydrant fuelling facility available in all stands.
2. Push back & Taxi procedure:
   - 2.1 Aircraft pushed back shall be towed till Tug Disconnect Point-4 before Engine start is carried out.

Apron 5				**Elevation: 26 FT**
Aircraft Stand No	Surface	PCR	Coordinates	Suitability
17	Concrete	990/R/A/W/T	100916.81N 0762355.38E	Code C
18	Concrete	990/R/A/W/T	100916.76N 0762354.04E	Code C
1. Hydrant fuelling facility available. 2. Taxi procedure: 2.1 Code C aircraft shall be towed forward till Tug Disconnect Point-5 before engine start is carried out. 2.2 Aircraft upto Code B may Power Out from stands.				

Apron 7					Elevation: 26 FT
Aircraft Stand No	Surface	PCR	Coordinates	Aircraft suitability (upto wingspan)	Helicopter suitability (overall dimension/max rotor diameter)
41	Concrete	890/R/A/W/T	100913.20N 0762409.92E	18 M	18M/14.5M
42	Concrete	890/R/A/W/T	100914.08N 0762409.88E	18 M	18M/14.5M
43	Concrete	890/R/A/W/T	100914.95N 0762409.85E	18 M	18M/14.5M
44	Concrete	900/R/A/W/T	100915.77N 0762409.82E	14 M	---
1. Stand Usage Procedure: 1.1 All stands are Power In. 1.2 For Power out from stands, aircraft shall be repositioned manually, facing West. 1.3 Hover turns not permitted with adjacent stands occupied. 2. Hydrant Fuelling facility not available.					

IAPP [ISOLATED AIRCRAFT PARKING POSITION]			Elevation: 26 FT	
Surface	PCR	Coordinates	Dimension	Suitability
Concrete	970/R/A/W/T	100913.39E 0762306.00N	60m radius	Code E
Hangar Apron				Elevation: 29 FT
Surface	PCR	Coordinates	Dimension	Suitability
Concrete	990/R/A/W/T	100924.24N 0762406.47E	---	Code C

5. Radar Surveillance coverage provided in Cochin Lower Area Control (LCI) with surveillance sensor inputs from Mangaluru, Bangalore (BIA) Radars and Cochin (ADS-B) are integrated in the ATS Automation System to enable the provision of Air Traffic Surveillance service for en route control.

Hours of operation of LCI notified through 'A' Series Notam.

6. Cochin ASMGCS Commissioned.

VOCI AD 2.24 CHARTS RELATED TO AN AERODROME

1. Aerodrome Chart (06 AUG 2026)
2. Aircraft Parking/Docking Chart (06 AUG 2026)
3. Aerodrome HOTSPOT (06 AUG 2026)
4. Aerodrome Obstacle Chart -Type A (Operating Limitations) RWY 09
5. Aerodrome Obstacle Chart - Type A (Operating Limitations) RWY 27
6. TORA from Different Runway Intersections - Departure RWY 09
7. TORA from Different Runway Intersections - Departure RWY 27
8. Locations of Different Exit Taxiways from Threshold - Arrival 09
9. Locations of Different Exit Taxiways from Threshold - Arrival 27
10. ILS Procudure RWY 09
11. VOR Procedure RWY 09
12. VOR Procedure RWY 27
13. ATC Surveillance Minimum Altitude Chart
14. ILS (Z) Procedure RWY 09 (17 April 2025)
15. Coding Table - ILS (Z) Procedure RWY 09 (17 April 2025)
16. Coding Table - RNP Procedure RWY 09 (17 April 2025)
17. RNP Procedure RWY 09 (17 April 2025)
18. Standard Departure Chart - Instrument (SID), RNAV1 (GNSS), RWY 09, (17 April 2025)
19. Standard Departure Chart - Instrument (SID), RNAV1 (GNSS), RWY 09, Tabular Description 1(17 April 2025)
20. Standard Departure Chart - Instrument (SID), RNAV1 (GNSS), RWY 09, Tabular Description 2 (17 April 2025)
21. Standard Departure Chart - Instrument (SID), RNAV1 (GNSS), RWY 09, Tabular Description 3 (17 April 2025)
22. Standard Departure Chart - Instrument (SID), RNAV1 (GNSS), RWY 27, (27 November 2025)
23. Standard Departure Chart - Instrument (SID), RNAV1 (GNSS), RWY 27, Tabular Description 1(17 April 2025)
24. Standard Departure Chart - Instrument (SID), RNAV1 (GNSS), RWY 27, Tabular Description 2 (27 November 2025)
25. Standard Departure Chart - Instrument (SID), RNAV1 (GNSS), RWY 27, Tabular Description 3 (17 April 2025)
26. Standard Arrival Chart - Instrument (STAR), RNAV1 (GNSS), RWY 09 (17 April 2025)
27. Standard Arrival Chart - Instrument (STAR), RNAV1 (GNSS), RWY 09, Tabular Description 1 (17 April 2025)
28. Standard Arrival Chart - Instrument (STAR), RNAV1 (GNSS), RWY 09, Tabular Description 2 (17 April 2025)
29. Standard Arrival Chart - Instrument (STAR), RNAV1 (GNSS), RWY 09, Tabular Description 3 (17 April 2025)
30. Standard Arrival Chart - Instrument (STAR), RNAV1 (GNSS), RWY 27 (17 April 2025)
31. Standard Arrival Chart - Instrument (STAR), RNAV1 (GNSS), RWY 27, Tabular Description 1 (17 April 2025)
32. Standard Arrival Chart - Instrument (STAR), RNAV1 (GNSS), RWY 27, Tabular Description 2 (27 November 2025)
33. Standard Arrival Chart - Instrument (STAR), RNAV1 (GNSS), RWY 27, Tabular Description 3 (17 April 2025)

VOCI AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	2	3